C9orf72-Nature2Agent

An autonomous AI scientist built from the C9orf72 ALS-autoimmunity study

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Abstract

This report describes a multi-agent system that converts the Michaelis et al. (2025) *Nature* study on C9orf72 autoimmunity in amyotrophic lateral sclerosis (ALS) into an interactive, executable AI scientist. The system ingests the paper, its supplementary material and its source data, exposes them as validated tools behind a Model Context Protocol (MCP) server, and runs a five-layer agent ecosystem that explains the design, reproduces the statistical analyses and figures, interrogates the peptide and HLA data, generates new hypotheses and verifies them. Reproduced p-values match the authors' reported values to within $|\Delta p| = 0.00131$. The discovery loop answered 24 questions and produced 8 verified observations that the paper does not state, among them a spatial clustering of epitope immunogenicity along the protein sequence (permutation $p = 2.00 \times 10^{-4}$) and the finding that the IL-5 bias in ALS is shared with TDP-43 while the IL-10 bias is specific to C9orf72. The system evaluates both the science (reproducibility, robustness, novelty, falsifiability) and itself (grounding, tool-use accuracy, determinism, rejection of planted false hypotheses), and frames a patient-stratified, antigen-specific therapeutic hypothesis with an explicit falsification experiment.

1 Introduction

Amyotrophic lateral sclerosis is a fatal neurodegenerative disease defined by the progressive loss of motor neurons. Neuroinflammation accompanies the degeneration, with T-cell infiltration and microglial activation in affected tissue, and T cells are thought to contribute to the pace of neuronal loss [1–3]. A specific antigenic target for this response had not been reported until Michaelis et al. identified C9orf72 as a target of ALS-associated T-cell autoreactivity and mapped the recognised epitopes [4]. Their central observations are that C9orf72-specific responses are increased in ALS, that the responses are mediated mainly by $CD4^+$ T cells, that they are relatively enriched for IL-5 and IL-10 over IFNγ, and that IL-10-mediated responses are associated with longer predicted survival.

The objective of this project is not to re-run that study. It is to ask whether an AI system can use the C9orf72 autoimmune mechanism as a starting point and move from observation to mechanism to hypothesis to a falsifiable therapeutic proposal, while remaining honest about what the data support. The work follows the Paper2Agent paradigm, which turns a paper and its computational resources into an executable agent served over MCP rather than treating the paper as a static document [5].

Contributions. The system (i) reconstructs the study's data as machine-readable tables and reproduces its headline claims from those tables; (ii) exposes the study as a validated MCP tool

server and a five-layer agent ecosystem; (iii) generates and verifies 8 observations absent from the paper; (iv) catalogues contradictions, uncertainties and limitations; and (v) evaluates both the scientific result and the AI system. All code was written with an AI coding agent, inspected, and covered by an automated test suite.

The pipeline the system is meant to demonstrate. The aim is not to show that a language model can summarise ALS literature. It is to show that a multi-agent system can move along a chain from an observation to a mechanism, from a mechanism to a hypothesis, from a hypothesis to a computational result, and from a result to a therapeutic candidate with a falsification experiment attached. Each link in that chain is a concrete artefact in this project: the observations are the reproductions, the mechanism is the IL-10-biased C9orf72-specific response, the hypotheses are the generated questions, the computational results are the verified observations, the candidate is the antigen-specific regulatory strategy, and the falsification is the experiment the designer agent specifies. The value of making the chain explicit is that every step can be checked, and a weak step is visible rather than hidden inside fluent prose.

2 Related work

Two lines of work frame this project. The first is the conversion of scientific artefacts into executable agents. Paper2Agent automates the construction of an MCP server from a paper and its codebase, with agents that locate the code, set up the environment, extract tools from tutorials and validate them before assembly [5]. The Michaelis study is an experimental immunology paper without a public analysis codebase, so the present work builds the computational infrastructure from the data rather than wrapping existing tutorials, then exposes it in the same MCP tool shape. The second line is multi-agent hypothesis generation, where separate agents propose, critique and debate hypotheses and a coordinating agent integrates them. The discovery loop here follows that division of labour, with explicit verification, replication, contradiction and falsification agents, and a principal-investigator agent that integrates without the authority to overrule evidence.

A recurring concern in both lines is the reliability of language-model output in a scientific setting, where a fluent but unsupported claim is worse than no claim. The design response taken here is to separate a deterministic computational core, which produces every number, from the language layer, which only frames and routes. In the committed configuration the framing layer is rule-based and reads from the computed results, so a figure in the text is always a figure that was computed. The evaluation in Section 9 measures this directly through a grounding check and a planted-false-hypothesis test.

3 Background and data

3.1 The study and its data

The study examined peripheral blood mononuclear cells from 40 people with ALS and 28 healthy controls, with additional Parkinson's disease (PD) and Alzheimer's disease (AD) cohorts and a validation cohort. Antigen-specific responses were measured by IFNγ, IL-5 and IL-10 FluoroSpot after a two-week expansion, against pools of C9orf72, TDP-43 and SOD1 peptides and an EBV control pool. Immunophenotyping used spectral flow cytometry, and cytokine-producing cells were characterised by intracellular cytokine staining. HLA class II binding was predicted with IEDB NetMHCIIpan-4.1 EL, and HLA restrictions were inferred with RATE [6, 7].

All primary data are open. The flow-cytometry files and FlowJo workspaces are deposited on Zenodo (CC-BY 4.0) across one spectral and four intracellular-cytokine-staining records; the full text, the figure source data and the supplementary tables are open access on PubMed Central (PMC12657233). The system fetches these with checksum verification and records provenance for each file. From them it builds fourteen tidy tables, including the complete 100-peptide screen, the per-donor HLA typing of 23 donors, and the per-donor source data behind every figure.

Why C9orf72 and why autoimmunity. A noncoding hexanucleotide repeat expansion in *C9orf72* is the most common genetic cause of ALS and frontotemporal dementia, and it carries a worse prognosis. The protein has roles in autophagy and membrane trafficking, and the expansion acts through a combination of reduced protein and toxic repeat products. Independently of the genetic mechanism, immune involvement is well documented: T cells infiltrate affected spinal cord, monocyte and mast-cell populations change in the motor-neuron niche [1, 2], and the number and suppressive function of regulatory T cells fall as the disease accelerates [8–10]. Against this background, a specific self-antigen recognised by autoreactive T cells gives the inflammation a target and a possible point of intervention, which is what the Michaelis study provides.

3.2 Machine-readable reconstruction

The source-data workbook stores each figure as several side-by-side panels, some of which reuse panel letters and measure names. A column-centric parser forward-fills the header rows and separates panels by column gaps, which prevents two different panels from being merged. The parse is cross-checked against the paper: epitope number 14 resolves to residues 56–70 (`DGEITFLANHTLNGE`), recognised by 7 of 22 donors with an average magnitude of 1216, which matches both the manuscript's Table 1 and the supplementary screen. The embedded C9orf72 sequence (UniProt Q96LT7, 481 residues) places all 95 non-phosphorylated peptides at their stated residue windows without error.

3.3 Acquisition and provenance

Acquisition is scripted and verified. The four intracellular-cytokine-staining Zenodo records are downloaded with their MD5 checksums and a per-record provenance manifest that stores the DOI, the license and the hash of every file; the full text and supplementary bundle are retrieved from the E-utilities and EuropePMC endpoints. Raw files are large and are not committed to version control; they are reproduced on demand by the fetch script, while the small derived tables are committed with attribution. Every processing step records a manifest with the git commit, the configuration hash, the package versions and the input checksums, so any table or figure can be traced to the exact state that produced it. Random seeds are fixed and dependencies are pinned to exact versions.

The reconstruction yields the thirteen analysis tables in Table 1, which together cover every figure's per-donor values, the complete peptide screen, the per-donor HLA typing and the two manuscript tables.

4 System architecture

The system is organised in five layers (Figure 1). Layer 1 is the scientific knowledge: the paper, its supplementary material, the source data, and curated external literature retrieved through a

Table 1: Machine-readable datasets reconstructed from the study's source data.

Dataset	Rows	Grain
`edfig2_ifng_il5_ratio`	792	per-donor cytokine
`edfig4_cd4_cd8`	456	per-donor CD4/CD8
`edfig5_immunogenicity_hla`	490	immunogenicity vs HLA
`edfig6_clinical_correlates`	403	per-donor clinical
`fig1_tcell_subsets`	819	per-donor subset
`fig2_fluorospot_responses`	782	per-donor FluoroSpot magnitude
`fig3_epitopes`	100	per-epitope
`fig4_mutation_survival`	284	per-donor by stratum
`reported_pvalues`	45	authors' p-values
`si_table1_peptide_screen`	100	per-peptide (100)
`si_table2_hla_typing`	506	per-donor x locus x allele
`table1_epitopes_ge3donors`	23	epitopes $\geq$3 donors
`table2_hla_restrictions`	7	HLA restrictions

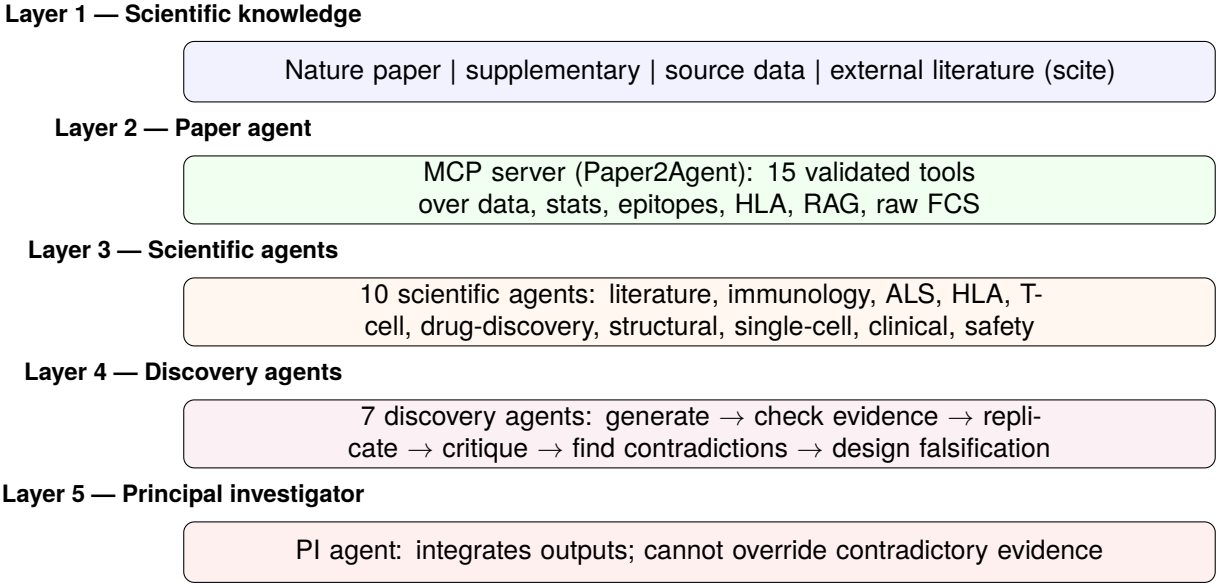


Figure 1: The five-layer agent ecosystem. Information flows upward from the data and tools to the scientific and discovery agents and finally to the PI agent. Every tool is shared between the MCP server and the agents, so a result produced interactively and a result produced by the discovery loop come from the same implementation.

literature agent. Layer 2 is the paper agent, an MCP server that exposes the study as fifteen validated tools. Layer 3 comprises ten scientific agents (literature, immunology, ALS, HLA, T-cell, drug-discovery, structural, single-cell, clinical and safety) that answer domain questions using those tools. Layer 4 comprises seven discovery agents (hypothesis generator, hypothesis critic, evidence checker, replication, contradiction, experiment designer and falsification). Layer 5 is a principal-investigator agent that integrates the outputs and, by construction, cannot override contradictory evidence.

4.1 Tools and reasoning backend

The computational core is a single tool registry shared by the MCP server and the agents. Tools cover design explanation, dataset access and querying, epitope and HLA interrogation, claim reproduction, named analyses, a generic group comparison, retrieval over the paper (a semantic RAG index with a lexical fallback), and a summary of the raw flow-cytometry files. Each tool

returns a JSON-serialisable result and is covered by the test suite, which plays the role of the Paper2Agent validation loop.

The reasoning agents use a pluggable backend. Without an API key they run deterministically: each answer is computed from the tools and the finding text is generated from the computed statistic, so no number is produced that is not grounded in a computation. With an API key the same agents run a Claude tool-use loop over the same tools. The committed results come from the deterministic backend, which makes the discovery loop reproducible without credentials. Every agent run is logged with its prompt, its tool calls and its output.

4.2 Statistics engine

The statistics follow the study's methods: two-tailed Mann–Whitney tests for two-group comparisons, one-way ANOVA with Dunnett's test for multi-group comparisons, Fisher's exact test for proportions, and Spearman's correlation, with Benjamini–Hochberg control of the false discovery rate and percentile bootstrap confidence intervals added for stress-testing. Effect sizes are reported as rank-biserial correlations alongside medians and sample sizes. The exploratory analyses add methods matched to each question. Spatial structure is tested with the lag-one autocorrelation of response frequency along the ordered tiling and a permutation null over 5000 shuffles. Donor dependence is assessed by a leave-one-out jackknife of the test statistic. Sensitivity to the assay floor is assessed by re-expressing the response as the proportion above the floor and testing it with Fisher's exact test. The discriminative value of a single feature is summarised by the area under the ROC curve. These choices let the system test not only whether an effect exists but whether it survives alternative assumptions.

5 Implementation

5.1 Tool catalogue

The MCP server exposes the fifteen tools in Table 2. They fall into four groups: orientation (design explanation, dataset listing and description), data access (filtered queries, reported-p-value retrieval, the protein sequence, a raw flow-cytometry summary), analysis (claim reproduction, named analyses, a generic group comparison, epitope and HLA interrogation), and retrieval (semantic search over the paper). A client such as Claude Code connects through the repository's MCP configuration and calls these tools interactively.

5.2 Scientific RAG

The paper's full text and supplementary information are chunked into 169 passages of about three sentences each. Retrieval combines a lexical BM25 score with cosine similarity over ONNX sentence embeddings, normalised and averaged; when the embedding model is unavailable the lexical score is used alone, so retrieval works offline. The Literature agent uses this index to ground statements in the paper's own wording, and the external literature is retrieved separately through the scite literature agent and stored with each reference's theme and relation to the work.

5.3 Raw flow-cytometry ingestion

Beyond the figure source data, the system reads the raw intracellular-cytokine-staining files directly. It indexes all 90 FCS files by group and condition, recovering the expected design of

Table 2: The fifteen tools exposed by the MCP server and used by the agents.

Tool	Purpose
`explain_experimental_design`	Summarise study design, cohorts, assays and statistical methods.
`list_datasets`	List available machine-readable datasets.
`describe_dataset`	Schema + head of a dataset.
`query_dataset`	Filter and return rows of a dataset.
`epitope_table`	Epitopes recognised by $\geq$ N donors.
`most_immunogenic_epitopes`	Most broadly immunogenic C9orf72 epitopes.
`broadest_hla_coverage`	Peptides with the broadest predicted HLA-II coverage.
`hla_typing`	Per-donor HLA typing (HLA-I and HLA-II).
`reproduce_claim`	Reproduce a headline claim of the paper.
`compare_groups`	Two-tailed Mann-Whitney between two groups.
`run_analysis`	Run a named analysis by id.
`list_analyses`	List all named analyses.
`reported_pvalues`	Authors' reported p-values (validation).
`protein_sequence`	C9orf72 protein sequence (UniProt Q96LT7).
`search_paper`	Retrieve passages from the paper + SI (scientific RAG).
`ics_fcs_summary`	Summarise the raw ICS flow-cytometry files.

roughly fifteen donors per group across the negative, peptide and positive conditions, for a total of 6.37 million recorded events across a fourteen-colour panel that includes a viability stain and lineage and cytokine markers. The FlowJo workspace gating hierarchy is read where parseable. This represents the primary data behind the study, not only the summarised figure values.

5.4 A peptide-immunogenicity model

As a model component, the system trains a classifier that predicts whether a peptide is immunogenic, defined as recognition by at least three donors, from its predicted HLA-II binding breadth and simple biochemical features of its sequence. On the 95 non-phosphorylated peptides (21 immunogenic), a cross-validated logistic model reaches an area under the ROC curve of 0.78, while gradient boosting overfits to 0.51. The simple linear model is therefore preferred on this small sample, which is itself a useful caution about model choice when the number of positive examples is in the low tens. The model is trained with a fixed seed and its checkpoint and metrics are saved.

5.5 Agent framework and the discovery loop

Each agent is a declarative specification: a name, a layer, a role, a system prompt and the subset of tools it may call. A recorder wraps the tool registry so that a tool call outside an agent's permitted set is refused and every permitted call is logged. The discovery loop proceeds in a fixed order. The hypothesis generator emits the question bank; for each question the owning scientific agent computes an answer from its tools; the evidence checker recomputes the answer and confirms it matches; the replication agent re-runs robustness checks where they apply; the critic and falsification agents attach rule-based weaknesses such as small samples or multiple-testing sensitivity; the contradiction agent classifies the result against the paper; and the principal-investigator agent selects the verified, novel, surviving results and compiles the catalogue of contradictions, uncertainties and limitations. The full agent roster is listed in Appendix A.

5.6 Design decisions

Three decisions shape the system. The first is a single shared tool registry behind both the MCP server and the agents, so that an interactive query and the autonomous discovery loop cannot disagree about what a tool computes. The second is a deterministic default backend, which trades the flexibility of free-text reasoning for the guarantee that every number in a result was computed rather than generated, with the live language-model backend kept available for when that flexibility is wanted. The third is to treat verification and criticism as first-class agents with their own logged runs, rather than as checks folded into the generator, so that the evidence for a result and the case against it are produced by different parts of the system and recorded separately.

5.7 Reproducibility and testing

The repository is organised around the directory structure required by the brief, with an installable library, pinned dependencies in both pip and conda forms, and a single entry point that runs the stages in order. The test suite covers the reproducibility core, the dataset parse, the statistics against the authors' values, the analyses, the figures, the tools, the MCP server, the RAG, the raw-FCS ingestion, the discovery loop and the evaluation. All AI-assisted code was read and is exercised by these tests.

6 Reproduction of the study's claims

The system reproduces every headline claim from the reconstructed data (Table 3). C9orf72-specific responses are higher in ALS than in controls for the total response ($p = 0.00159$), for IL-5 ($p = 3.18 \times 10^{-4}$), for IL-10 ($p = 0.00469$) and, more weakly, for IFNγ ($p = 0.0381$); the pattern is shown in Figure 2. The IFNγ/IL-5 ratio is lower in ALS, which is the IL-5 bias the study reports. C9orf72-reactive cells are predominantly CD4$^+$ ($p \approx 3 \times 10^{-6}$). Mutation carriers respond more strongly, IL-10 responses are higher in donors with longer predicted survival, and predicted HLA-II binding breadth correlates with observed immunogenicity.

As an external check, the system compares its recomputed p-values against the authors' own reported values for the Extended Data Figure 2 comparisons. The two agree in every significance call and differ by at most $|\Delta p| = 0.00131$, a gap consistent with the difference between an exact test and the normal approximation with tie correction.

The survival and mutation strata reproduce as well (Figure 3). IL-10 responses are higher in donors with longer predicted survival, and the total response is higher in C9orf72 mutation carriers than in non-carriers. Both strata have small per-group samples, so these effects are individually marginal even where their direction matches the paper, which is recorded as an uncertainty rather than smoothed over.

7 Epitope and HLA interrogation

The agent ranks the 100 screened peptides by breadth and magnitude and returns the HLA coverage of each. The most immunogenic epitope is residues 56–70 (Table 4), which is also the peptide with the broadest predicted HLA-II coverage. Immunogenicity across the protein is not flat: it rises and falls in runs along the sequence (Figure 4), a structure examined quantitatively in the next section. The inferred HLA restrictions reproduce the study's Table 2, pairing specific peptides with DRB1, DQB1 and related alleles (Table 5).

Table 3: Reproduction of the study's headline claims from the reconstructed source data.

Question	p	Paper claim reproduced
Are C9orf72 Total FluoroSpot responses different in	0.00159	C9orf72-specific responses increas
Are C9orf72 IFNg FluoroSpot responses different in A	0.0381	C9orf72-specific responses increas
Are C9orf72 IL-5 FluoroSpot responses different in A	3.18×10^{-4}	C9orf72-specific responses increas
Are C9orf72 IL-10 FluoroSpot responses different in	0.00469	C9orf72-specific responses increas
Do ALS responses show an IL-5/IL-10 (Th2/regulatory)	0.0483	ALS responses enriched for IL-5/IL
Is the ALS Total C9orf72 response specific vs PD and	–	C9orf72 response relatively specif
Do C9orf72 mutation carriers show stronger Total C9o	0.0281	Mutation carriers show stronger C9
Is the IL-10 C9orf72 response associated with predic	0.0382	IL-10-mediated responses associate
Is predicted HLA-II binding breadth associated with	0.00385	Immunogenicity associated with HLA
Are C9orf72-reactive T cells predominantly CD4+?	–	Responses predominantly CD4+ T-cel

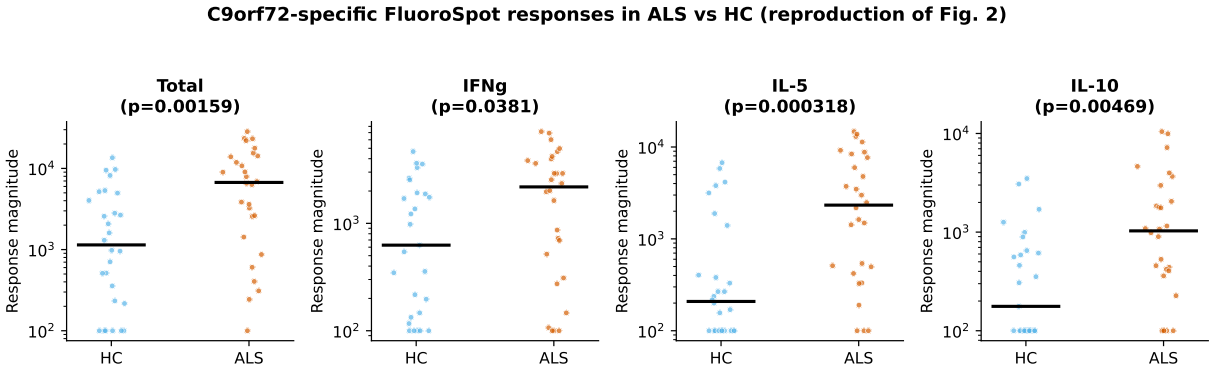


Figure 2: C9orf72-specific FluoroSpot responses in ALS versus healthy controls, by cytokine (reproduction of the study's Figure 2). Each point is one donor; the black bar is the median; the y-axis is on a log scale. Healthy controls are shown in sky blue and ALS donors in vermilion. The p-value above each panel is a two-tailed Mann–Whitney test.

The relationship between predicted HLA-II binding and observed immunogenicity holds across the full screen (Figure 5). Peptides predicted to bind more of the common class-II alleles tend to be recognised by more donors (Spearman $\rho = 0.29$, $p = 3.85 \times 10^{-3}$, 95 peptides). The association is clear but far from deterministic, which is expected because recognition depends on antigen processing and the T-cell repertoire, not on binding alone.

7.1 Answering the study's open questions

The brief poses five questions that an agent built from this study should be able to answer. The system answers each from the data. The most broadly immunogenic epitopes are residues 56–70 (recognised by 7 of 22 donors), 366–380 (6 of 20) and 426–440 (3 of 19 but with high magnitude); the first of these also has the broadest predicted HLA-II coverage, with 27 predicted binders among the common class-II alleles. The cytokine signatures that distinguish response phenotypes are IL-5 and IL-10 against IFNγ, with the refinement developed in the next section

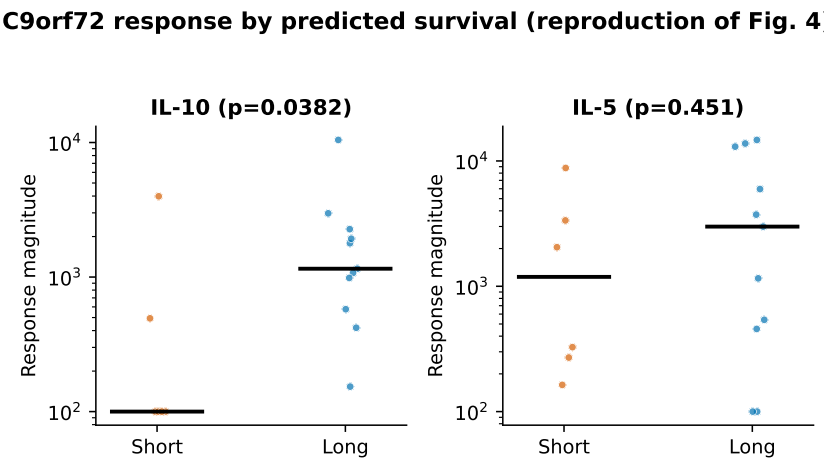


Figure 3: C9orf72 response by predicted survival (reproduction of part of Figure 4). IL-10 and IL-5 response magnitudes in donors with short versus long predicted survival; the black bar is the median and the y-axis is on a log scale.

Table 4: The eight most immunogenic C9orf72 peptides by average magnitude, with response frequency and the number of predicted HLA-II binders.

#	Residues	Sequence	Resp./Tested	Mag.	HLA
14	56-70	DGEITFLANHTLNGE	7/22	1216	27
88	426-440	PFKSLRNLKIDLDLT	3/19	871	10
22	96-110	SLIFDGNWNGDRSTY	1/20	868	3
87	421-435	QKGKKPFKSLRNLKI	4/19	769	20
91	431-445	RNLKIDLDLT*AEGDL	5/19	718	
37	171-185	SIIPMLTGEVIPVME	1/27	697	4
55	261-275	KCSRLCEAESSFKYE	2/25	692	0
52	246-260	KIVRTLCLFLTPAER	2/25	682	0

that the IL-10 component is specific to C9orf72 while the IL-5 component is shared with TDP-43. The observations that depend on particular donors are the survival and mutation strata, where per-group samples are small, whereas the central ALS-versus-control total response does not depend on any single donor. The conclusions that remain robust under alternative statistical models are the total, IL-5 and IL-10 increases, which survive false-discovery correction, while the IFNγ increase does not, and the total increase is carried by the magnitude distribution rather than by the fraction of donors that respond.

8 New observations

The discovery loop generated 24 questions owned by the scientific agents, computed an answer for each, re-verified each answer independently, applied an adversarial critique, and classified each result against the paper. Table 7 lists the 8 observations that are novel relative to the paper and that survived verification. Several are worth describing.

Immunogenicity is spatially clustered. Ordering the peptides along the protein, the response frequency of adjacent overlapping peptides is positively autocorrelated at lag one ($\rho = 0.47$, permutation $p = 2.00 \times 10^{-4}$ over 5000 permutations), shown in Figure 6. Immunogenic residues occur in contiguous stretches rather than at isolated positions, which is consistent with a small number of antigenic regions each spanning several overlapping peptides. The paper maps indi-

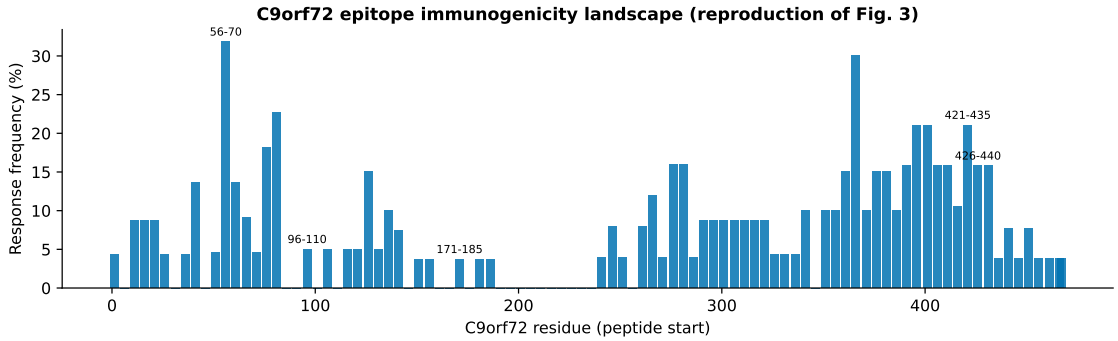


Figure 4: Immunogenicity along the C9orf72 sequence (reproduction of the study's Figure 3). Each bar is the response frequency of one 15-mer peptide at its start residue; the five peptides with the highest average magnitude are labelled by residue range.

Table 5: HLA-restricted C9orf72 peptides and corresponding alleles (reproduction of Table 2).

Peptide sequence	Allele	RATE P value	Binding percentile ranka
VRHIWAPKTEQVLLS	DRB1*14:01	0.025	29.0
TPDLNIFQDVLHRDT	DRB1*01:02	0.022	20.0
LDQVFQLKPGLSLRS	DRB1*01:02	0.022	3.3
IFQDVLHRDTLVKAF	DRB1*03:01	0.028	4.9
HRKALTLIKYIEDDT	DQB1*05:01	0.05	22.0
PFKSLRNLKIDLDLT	DQB1*05:01	0.05	11.0
RNLKIDLDLTAEGDL	DQB1*05:01	0.05	12.0

vidual epitopes but does not report this spatial structure.

The cytokine bias is antigen-dependent. The study frames the IL-5 and IL-10 bias around C9orf72. Testing each antigen separately (Figure 7), the ALS increase in IL-5 is significant for C9orf72 and also for TDP-43, so the IL-5 skew is not unique to C9orf72. The IL-10 increase, by contrast, is significant only for C9orf72. This distinction matters for interpretation: a shared IL-5 response may reflect a general ALS T-helper-2 skew, whereas the IL-10 signal is antigen-specific and is the component that tracks survival.

Robustness and its limits. The central C9orf72 total-response difference is robust to removing any single donor: across all 56 leave-one-out removals the p-value stays below 0.05. Two robustness limits are also surfaced by the system itself. First, if the total response is reduced to the fraction of donors above the assay floor, the difference weakens to Fisher $p = 0.10$, so the effect lives in the magnitude distribution rather than in the mere presence of a response. Second, the IFNγ increase does not survive Benjamini–Hochberg correction across the sixteen antigen-by-cytokine comparisons ($q = 0.12$), whereas the total, IL-5 and IL-10 effects do. These are reported as limitations, not hidden.

Multiple-testing and classifier views. Two further results quantify how far the signal carries. Under Benjamini–Hochberg control across the sixteen antigen-by-cytokine comparisons, the total, IL-5 and IL-10 C9orf72 effects keep $q < 0.05$ while IFNγ does not, which is consistent with the IL-5/IL-10 emphasis of the study. Treating the IFNγ/IL-5 ratio as a single feature, its ability to separate ALS from healthy controls is modest (area under the ROC curve 0.68), so the cytokine bias is a real group difference rather than an individual-level diagnostic.

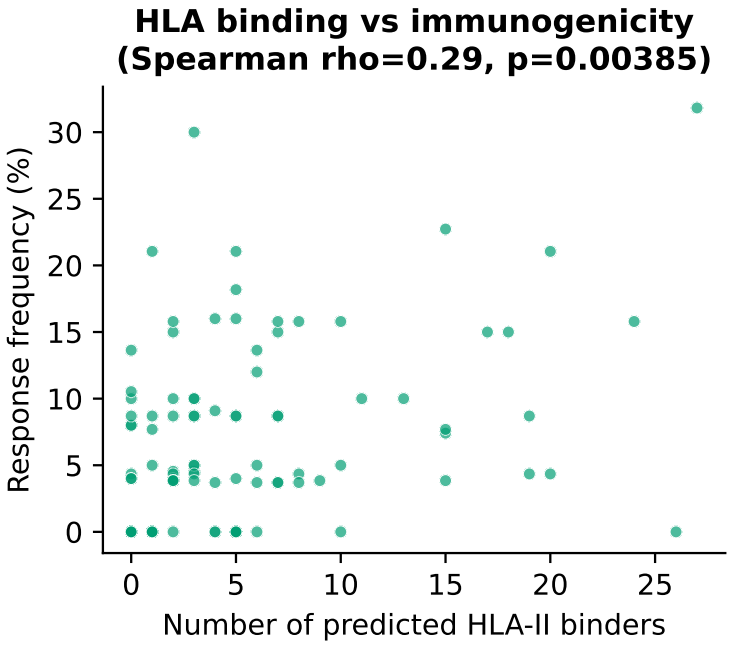


Figure 5: Predicted HLA-II binding breadth against observed response frequency across the 95 non-phosphorylated peptides (reproduction of Extended Data Figure 5). Each point is one peptide.

Phosphorylation. The screen includes five phosphorylated peptide variants alongside their parent sequences. Comparing each pair, phosphorylation changed the average magnitude by a median of about nine spot units and raised immunogenicity in three of the five pairs. The sample is small and the effect is weak, which the critic agent flags; the result is reported as a direction to test rather than a conclusion.

The full question set, with its classification against the paper, is listed in Table 6.

9 Evaluation

The system evaluates the scientific result and itself (Table 8). On the scientific side it reproduces 10 of 10 reproduction questions, agrees with the authors' reported p-values in every significance call, and meets the success criterion of at least five verified novel observations. On the AI side the grounding rate is 1.00, because every reported number is generated from a computed statistic rather than asserted; tool-use accuracy is 1.00; the discovery loop is deterministic across repeated runs; and when two false hypotheses are planted (an EBV control difference and a donor-ordering artefact), the system rejects both (2/2). The system also flags its own weak results: ten questions carry explicit caveats such as small samples, marginal significance or multiple-testing sensitivity.

The AI metrics are defined operationally. Grounding is the fraction of quantitative findings whose reported number matches the computed statistic, so a value of one means no number was invented. Tool-use accuracy is the fraction of tools that return a valid result when called with default arguments. Determinism is checked by running the discovery loop twice and comparing the observation set. Self-error detection counts the questions the system annotates with a weakness, and the false-hypothesis-rejection test plants two nulls, a control-antigen group difference and a donor-ordering artefact, that a sound system must fail to confirm. The scientific metrics follow the brief: reproducibility against the paper, robustness under alternative assumptions, novelty as the count of verified new observations, and falsifiability as the presence of an explicit test for each

Table 6: The 24 questions generated and answered by the discovery loop.

ID	Kind	Question	vs paper
Q01	reproduction	Are C9orf72 total FluoroSpot responses increased in ALS vs HC?	agrees
Q02	reproduction	Are C9orf72 IL-5 responses increased in ALS vs HC?	agrees
Q03	reproduction	Are C9orf72 IL-10 responses increased in ALS vs HC?	agrees
Q04	reproduction	Are C9orf72 IFNg responses increased in ALS vs HC?	agrees
Q05	reproduction	Do ALS responses show an IL-5/IL-10 bias over IFNg?	agrees
Q06	reproduction	Are C9orf72-reactive T cells predominantly CD4+?	agrees
Q07	reproduction	Is the ALS C9orf72 response specific vs PD and AD controls?	agrees
Q08	reproduction	Do C9orf72-mutation carriers show stronger responses?	agrees
Q09	reproduction	Is the IL-10 C9orf72 response associated with predicted survival?	agrees
Q10	reproduction	Is predicted HLA-II binding associated with observed immunogenicity?	agrees
Q11	extension	Which C9orf72 epitopes have the broadest predicted HLA-II coverage?	neutral
Q12	extension	Which C9orf72 epitopes are most broadly immunogenic?	neutral
Q13	novel	Is the ALS IL-5 bias specific to C9orf72 or shared across neuroantigen	caveats
Q14	novel	Is the ALS IL-10 bias specific to C9orf72 or shared across neuroantige	extends
Q15	novel	Do phosphorylated peptide variants change immunogenicity?	extends
Q16	novel	Is epitope immunogenicity spatially clustered along the sequence?	extends
Q17	novel	Does the central ALS-vs-HC result depend on any single donor?	extends
Q18	novel	Is the ALS-vs-HC difference robust to the assay floor (censoring)?	caveats
Q19	novel	Do the C9orf72 differences survive family-wise FDR correction?	caveats
Q20	novel	How well does the IFNg/IL-5 ratio alone separate ALS from HC?	extends
Q21	interpretive	What antigen-specific therapeutic strategy does the mechanism suggest?	neutral
Q22	interpretive	What experiment would most efficiently falsify the core hypothesis?	neutral
Q23	interpretive	What single-cell follow-up would resolve the CD4 phenotype identity?	neutral
Q24	interpretive	Is the C9orf72 autoimmunity finding supported by external literature?	neutral

hypothesis.

10 From observation to therapeutic hypothesis

The drug-discovery and experiment-designer agents turn the mechanism into a testable proposal. The reasoning is as follows. IL-10-biased C9orf72 responses track longer predicted survival, and IL-10 from regulatory T cells is associated with neuroprotection in several settings [11–13]; regulatory T-cell number and function also decline as ALS progresses [8–10]. These threads suggest a tolerogenic, antigen-specific strategy: expand or induce regulatory C9orf72-specific T cells against the dominant epitopes, stratified to mutation carriers who show the strongest responses, with the broadest HLA-covering epitopes as the most deployable antigens. An antigen-specific regulatory approach has been pursued in ALS and in autoimmune disease more generally [14, 15], though a recent polyclonal regulatory T-cell trial in a different autoimmune indication was negative, which is a reason for caution rather than optimism [16].

The experiment-designer agent states the most efficient falsification: expand C9orf72(56–70)-specific CD4 T cells, polarise them toward an IL-10-producing regulatory phenotype, and test adoptive transfer in a C9orf72 model, with the pre-registered falsifier that if IL-10-biased C9orf72-specific regulatory T cells do not slow progression, the survival-association hypothesis is rejected. The single-cell agent proposes the companion experiment needed to interpret the result: single-cell transcriptomics with paired T-cell-receptor sequencing on sorted C9orf72-reactive CD4 cells, to resolve whether the IL-10 signal comes from a Tr1/regulatory or a T-helper-2 effector population. These are hypotheses with explicit tests, not claims of efficacy.

Table 7: Verified observations not explicitly reported in the paper. “caveats” marks a robustness limitation the system surfaced about the paper’s framing; “extends” marks a new result consistent with the paper.

ID	Verified novel observation	vs paper
Q13	ALS>HC IL-5 enrichment is significant for: C9orf72, TDP-43. Not unique to C9orf72.	caveats
Q14	ALS>HC IL-10 enrichment is significant for: C9orf72. Consistent with C9orf72 focus.	extends
Q15	Across 5 phospho/parent pairs, phosphorylation changed average magnitude by a median of +8.7 spot units; 3/5	extends
Q16	Adjacent overlapping peptides show lag-1 autocorrelation of response frequency ρ=0.47 (permutation p=0.000	extends
Q17	Base p=0.00159; across all leave-one-out removals p stays in [0.000607, 0.00263] and remains $<$0.05 in 56/56	extends
Q18	Rank-based p=0.00159; above-floor responders 27/28 (ALS) vs 22/28 (HC), Fisher p=0.101: weaker under censori	caveats
Q19	After BH-FDR over all 16 antigen x cytokine comparisons, 3/4 C9orf72 cytokines remain significant (q$<$0.05):	caveats
Q20	The IFNg/IL-5 ratio alone separates ALS from HC with AUC=0.68 (n_ALS=26, n_HC=17): a modest single-feature s	extends

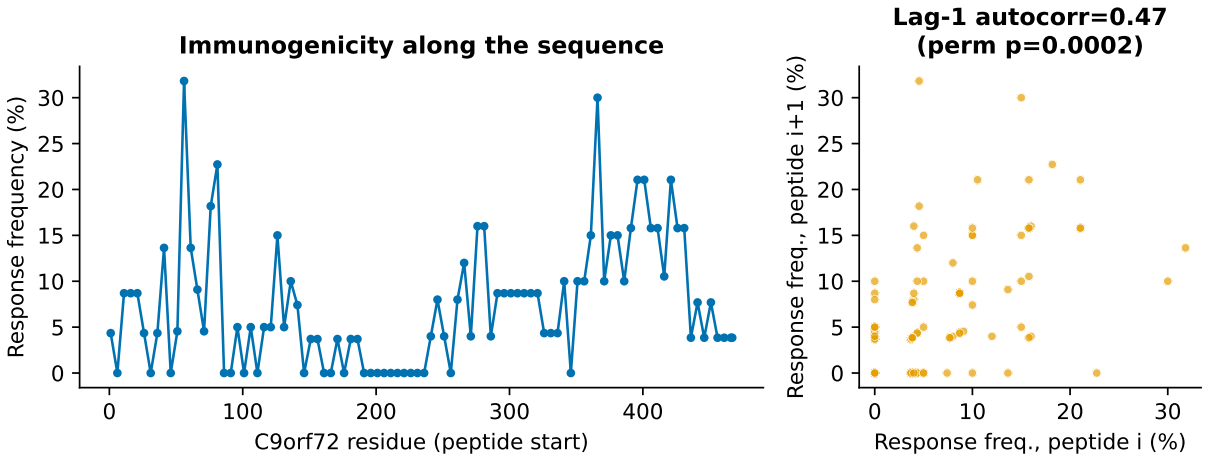


Figure 6: Spatial structure of immunogenicity. Left: response frequency along the C9orf72 sequence. Right: response frequency of each peptide against that of the next peptide in the tiling; the positive lag-1 autocorrelation indicates clustering.

11 Limitations

The system inherits the study’s limits and adds its own. FluoroSpot magnitude is in assay-specific spot units with a floor, and the total-response difference depends on the magnitude distribution rather than on response presence alone. The IFNγ increase does not survive multiple-testing correction. Survival and mutation strata have small per-group samples, so those effects are individually marginal even where consistent with the paper. HLA-II binding uses the authors’ NetMHCIIpan predictions rather than a fresh prediction run. The reasoning agents, in the committed configuration, do not call an external language model, so the generative and critical steps are rule-based and grounded in the tools; the live backend is available but was not used for the recorded results. None of these undermine the reproductions, and each is catalogued with the result it qualifies.

The agentic approach has its own limits that are worth stating plainly. A deterministic generator explores the hypothesis space that its rules and question bank define, so its breadth is bounded by what was anticipated; a live language-model generator would widen that space at the cost of the

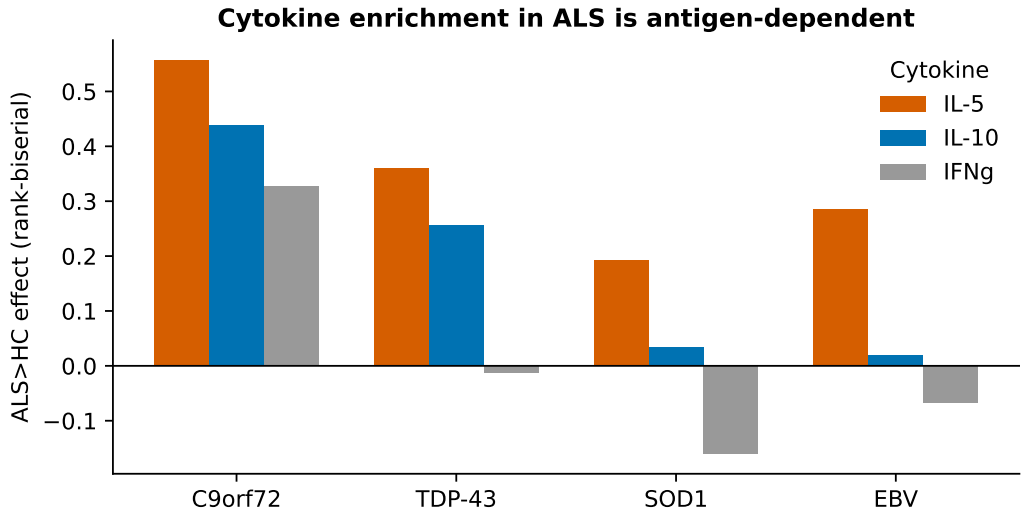


Figure 7: The ALS-versus-HC cytokine effect by antigen, as a rank-biserial effect size (positive means higher in ALS). The IL-5 increase is present for both C9orf72 and TDP-43, whereas the IL-10 increase is specific to C9orf72.

Table 8: Scientific and AI-system evaluation.

Metric	Value
Paper claims reproduced	10/10
Reported-p fidelity (max $\|\Delta p\|$)	0.00131
Verified novel observations	8
Grounding rate (no hallucinated numbers)	1.00
Tool-use accuracy	1.00
Deterministic reproduction	True
Planted nulls correctly rejected	2/2
Questions self-flagged with caveats	10

grounding guarantee, which is the trade this project resolves in favour of grounding. The critique is rule-based and catches the common failure modes, small samples, marginal significance and multiple testing, but it does not reason about confounds it was not told to look for. The therapeutic proposal is a hypothesis assembled from the data and the literature; it is not evidence of efficacy, and the system is built to make that boundary explicit rather than to cross it. These are reasons to read the output as a structured, checkable argument, not as a conclusion.

A mechanistic reading. The study's observations fit a coherent, if still tentative, mechanism. A loss of peripheral tolerance to C9orf72 would expose the protein to autoreactive CD4 T cells, which is consistent with the CD4 dominance of the response and with the stronger responses in mutation carriers, in whom C9orf72 handling is already perturbed. The quality of that response then appears to matter for outcome. An IFNγ-dominated response is the signature of a pro-inflammatory, potentially tissue-damaging T-helper-1 programme, whereas IL-10 is the central regulatory cytokine of suppressive T cells and is linked to neuroprotection through the dampening of microglial activation in several models [11–13]. The finding that the response in ALS is shifted away from IFNγ and toward IL-10, and that the IL-10 share tracks longer predicted survival, is therefore readable as a protective regulatory arm that is present to differing degrees across patients. The decline of regulatory T cells as ALS progresses [8, 9] places this arm under strain exactly when it is most needed. The intervention that follows from this reading is to reinforce the protective arm rather than to suppress the response wholesale, which is why the proposed strategy is tolerogenic and antigen-specific rather than broadly immunosuppressive.

Why stratification matters here. The data argue for stratification on two axes. Mutation carriers show the strongest C9orf72-specific responses, so they are the subgroup in whom an antigen-specific intervention has the most to act on. Predicted survival separates donors whose response is IL-10-biased from those whose response is not, and the IL-10-biased group is the one with the better outcome, which suggests that the regulatory character of the response, rather than its mere size, is the feature to reinforce. The antigen-level result sharpens this: the IL-10 signal that tracks survival is specific to C9orf72, whereas the IL-5 signal is shared with TDP-43, so an intervention aimed at the protective component should be directed at the C9orf72-specific, IL-10-producing population rather than at the broader T-helper-2 response. A therapy that expands protective regulatory cells carries a risk in the other direction, that antigen-specific tolerance blunts a response that is itself partly protective, which is why the safety agent keeps this hazard on the list and why the proposed first test is a falsification rather than a dose-finding study.

12 Future work

Three extensions follow naturally. The first is to run the reasoning layer live over the same tools and to measure the gap between the deterministic framing and a language-model framing on the grounding and tool-use metrics, which would quantify what the language layer adds and where it drifts. The second is to re-derive the response frequencies from the raw flow-cytometry files by applying the FlowJo gating, which would let the system test the figures against an independent quantification rather than the authors' summarised values, and would extend the novel-observation search to the single-cell phenotype. The third is to connect this paper agent to agents built from complementary papers, so that a hypothesis generated here can be checked against external datasets and models, which is the cross-project ecosystem the programme envisages.

13 Conclusions

The project shows that a paper and its data can be turned into an interactive AI scientist that does more than summarise. The system reproduces the study's claims from its own data with high fidelity, produces eight verified observations the paper does not state, surfaces two genuine robustness caveats about the original framing, and frames a falsifiable therapeutic hypothesis with a companion experiment. Every result is computed, logged and re-verified, and the system is honest about what it cannot support. The same tools that answer a question interactively drive the discovery loop, so the interactive agent and the autonomous pipeline agree by construction.

Reproducibility and AI use. All data fetching is checksum-verified, dependencies are pinned, seeds are fixed, and each run writes a provenance manifest. The AI tools used, the agent specifications and prompts, and the record of hypotheses, evidence and rejected claims are documented in the repository (`AI_USAGE.md`, `agents/registry.json`, `results/discovery/`). The author inspected and tested all AI-assisted code and is responsible for the scientific content.

A Agent roster

The eighteen agents across the three operational layers are listed in Table 9. Each is defined by a specification with a system prompt and a permitted tool set, exported to `agents/registry.json`, and every run is logged with its prompt, tool calls and output.

Table 9: The agent roster (Layers 3–5).

Agent	Layer	Role
literature agent	Scientific	Find external support/contradiction for findings.
immunology agent	Scientific	Interpret cytokine responses and Th phenotype.
als agent	Scientific	Relate findings to ALS biology, mutation status and contro
hla agent	Scientific	Analyse HLA typing, restriction and binding coverage.
tcell agent	Scientific	Analyse CD4/CD8 and T-cell subset composition.
drug discovery agent	Scientific	Propose antigen-specific therapeutic strategies.
structural agent	Scientific	Analyse epitope sequence/position and modifications.
singlecell agent	Scientific	Propose single-cell follow-ups for phenotype identity.
clinical agent	Scientific	Relate responses to survival and clinical variables.
safety agent	Scientific	Stress-test robustness, multiple testing and censoring.
hypothesis generator	Discovery	Generate testable questions from the data.
hypothesis critic	Discovery	Adversarially critique each hypothesis/answer.
evidence checker	Discovery	Independently recompute and verify each finding.
replication agent	Discovery	Re-run robustness checks (jackknife, FDR, censoring).
contradiction agent	Discovery	Classify each finding vs the paper's claims.
experiment designer	Discovery	Design the most efficient falsification experiment.
falsification agent	Discovery	State what observation would falsify each claim.
pi agent	PI	Integrate all outputs; cannot override contradictory evide

B Contradictions, uncertainties and limitations

The principal-investigator agent compiles the catalogue in Table 10. Items marked as caveats qualify the paper's framing; items marked as limitations or uncertainties qualify the present analyses.

Table 10: The catalogue produced by the discovery loop.

ID	Type	Item
Q13	caveat	ALS>HC IL-5 enrichment is significant for: C9orf72, TDP-43. Not unique to C9orf72.
Q18	caveat	Rank-based p=0.00159; above-floor responders 27/28 (ALS) vs 22/28 (HC), Fisher p=0.101: weaker under censoring.
Q19	caveat	After BH-FDR over all 16 antigen x cytokine comparisons, 3/4 C9orf72 cytokines remain significant ($q<0.05$): Total q=0.0127, IFNg q=0.122, IL
L1	limitation	FluoroSpot magnitude is in assay-specific spot units with a floor of 100; the ALS>HC total-response difference weakens when reduced to the f
L2	limitation	The C9orf72 IFNg ALS>HC increase does not survive BH-FDR across the 16-comparison family (q=0.12), unlike IL-5/IL-10/Total.
L3	uncertainty	Survival and mutation strata have small per-group n; effects are consistent with the paper but individually marginal.

C Reproducibility checklist

Fixed random seeds (`configs/base.yaml`); exact dependency pins (`requirements.txt`, `environment.yml`); data versioned by Zenodo record identifier and per-file MD5; per-run provenance manifests (git commit, configuration hash, package versions, input checksums); deterministic discovery loop verified by re-running; logged agent runs; recorded prompts and agent specifications; and executable stage scripts driven by a single entry point. The AI tools used and the record of hypotheses, evidence and rejected claims are documented in `AI_USAGE.md` and under `results/discovery/`.

D Methods in detail

D.1 Source-data parser

The source-data workbook stores each figure as side-by-side panels marked a, b, c and so on, with one or more header rows above columns of per-donor values. The parser forward-fills each header row across the full width, then walks the columns left to right. A run of columns separated from the previous run by a blank column starts a new block, and the block index is attached to every column. This block index is what separates two panels that reuse the same letter or the same measure name, such as the two-group and four-group blocks of Extended Data Figure 2, which a parser keyed on the measure name alone would merge. Each emitted column carries its block, its panel letter, its measure and its series label, and the numeric cells beneath it, which the per-figure builders then map to semantic fields.

D.2 Statistical definitions

Two-group comparisons use the two-tailed Mann–Whitney U test, with the rank-biserial correlation $r = 1 - 2U/(n_1 n_2)$ as the effect size, where a negative value under the convention used here means the first group ranks higher. Proportions use Fisher's exact test and correlations use Spearman's ρ. Multi-group comparisons use one-way ANOVA with Dunnett's test against the control group. The false discovery rate across a family of tests is controlled by the Benjamini–Hochberg procedure, reported as adjusted q-values. Spatial structure uses the lag-one autocorrelation of response frequency along the ordered tiling, with a two-sided permutation p-value from 5000 shuffles. Donor dependence uses a leave-one-out jackknife of the test statistic, and sensitivity to the assay floor compares the fraction of donors above the floor with Fisher's exact test.

D.3 Discovery procedure

The loop runs in a fixed order. The hypothesis generator emits the question bank. For each question, the owning scientific agent computes an answer through its permitted tools. The evidence checker recomputes the answer from scratch and compares the serialised results. The replication agent runs the relevant robustness analysis where one applies. The critic and falsification agents attach rule-based flags for small samples, marginal significance, underpowered pairings and single-feature weakness. The contradiction agent classifies the result as agreeing with, extending or qualifying the paper. The principal-investigator agent keeps the results that are novel, verified and free of a disqualifying flag as the observation set, and compiles the catalogue of caveats, uncertainties and limitations. Every step writes a run record with its prompt, its tool calls and its output.

D.4 Evaluation metrics

Grounding is the fraction of quantitative findings whose printed number equals the computed statistic. Tool-use accuracy is the fraction of tools returning a valid result under default arguments. Determinism is equality of the observation set across two runs. Self-error detection is the count of questions carrying at least one critique flag. The false-hypothesis-rejection test reports the fraction of planted nulls that the system fails to confirm. Reproduction fidelity is the maximum absolute difference between a reproduced and a reported p-value over the comparisons for which the authors publish a value, together with the agreement of the significance calls.

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